

Session 1

Q1. Name 5 IT job roles and write one line about what each role does.

Ans: 1. **Software Developer** – Designs, develops, and maintains software applications and programs.

2. **Web Developer** – Creates and maintains websites and web applications using programming languages.

3. **Database Administrator (DBA)** – Manages, secures, and maintains databases to ensure data is stored safely and efficiently.

4. **Network Engineer** – Installs, configures, and maintains computer networks to ensure smooth communication between devices.

5. **Cybersecurity Analyst** – Protects computer systems and networks from cyber threats, hackers, and security breaches.

Q2. List 3 product companies and 3 service companies with one example of work each does.

Ans: **Product Companies:**

1. **Apple** – Develops and sells products like the iPhone, MacBook, and iPad.

2. **Microsoft** – Develops software products such as Windows operating system and Microsoft Office.

3. **Google** – Develops products like Android, Google Chrome, and Pixel smartphones.

Service Companies:

1. **Tata Consultancy Services (TCS)** – Provides IT consulting and software development services to businesses.
2. **Infosys** – Offers software development, cloud computing, and digital transformation services.
3. **Wipro** – Provides IT services, cybersecurity, and business consulting solutions.

Q3. Write the full forms and one-line use of: GitHub, Slack, Jira, IDE, API.

Ans: 1. **GitHub** – *Git Hub*

Use: A platform used to store, manage, and collaborate on software code using Git.

2. **Slack** – *Searchable Log of All Conversation and Knowledge*

Use: A communication tool used by teams for messaging, file sharing, and collaboration.

3. **Jira** – *it has no official full form*

Use: A project management tool used to track tasks, bugs, and software development progress.

4. **IDE** – *Integrated Development Environment*

Use: A software application that helps programmers write, edit, run, and debug code.

5. **API** – *Application Programming Interface*

Use: A set of rules that allows different software applications to communicate with each other.

Q4. Name any 5 programming languages and mention one popular use case for each.

Ans: 1. **Python** – Used for Artificial Intelligence (AI), Machine Learning, and data analysis.

2. **Java** – Used for developing Android applications and enterprise software.

3. **C++** – Used for game development and high-performance applications.

4. **JavaScript** – Used for creating interactive websites and web applications.

5. **PHP** – Used for developing dynamic websites and web applications.

Q5. What is the difference between a developer and a tester? Answer in 2 lines.

Ans: 1. A **developer** writes and develops software or applications by creating code.

2. A **tester** checks the software for bugs and errors to ensure it works correctly before release.

Session 2

Q1. What are the three main layers of software architecture? Write one line about each.

Ans: 1. **Presentation Layer (UI Layer)** – It is the user interface where users interact with the application.

2. **Business Logic Layer (Application Layer)** – It processes data and implements the application's business rules and logic.

3. **Data Layer (Database Layer)** – It stores, retrieves, and manages data in the database.

Q2. Define client and server in your own words with one real-life example.

Ans: **Client:** A client is a device or application that sends a request to access data or services from a server.

Server: A server is a computer or system that receives requests from clients and provides the requested data or services.

Real-life Example: When you open YouTube on your phone, your phone acts as the **client**, and YouTube's computers act as the **server**, sending videos to your device.

Q3. Write 2 differences each between desktop apps, web apps, and mobile apps.

Ans: **Desktop Apps**

1. Installed directly on a computer or laptop.
2. Can work without an internet connection (for many applications).

Web Apps

1. Run in a web browser and do not require installation.
2. Usually require an internet connection to access.

Mobile Apps

1. Installed on smartphones or tablets through an app store.
2. Designed for touch-screen devices and mobile operating systems like Android or iOS.

Q4. Name any 3 databases and mention whether each is SQL or NoSQL.

Ans: 1. **MySQL – SQL** database.

2. **PostgreSQL – SQL** database.

3. **MongoDB – NoSQL** database.

Session 3

Q1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

Ans: 1. **HTTP – HyperText Transfer Protocol**

Use: It is used to transfer web pages and data between a web browser and a web server.

2. **HTTPS – HyperText Transfer Protocol Secure**

Use: It securely transfers data between a web browser and a web server using encryption.

3. **FTP – File Transfer Protocol**

Use: It is used to upload and download files between a computer and a server.

4. **SMTP – Simple Mail Transfer Protocol**

Use: It is used to send emails from one mail server to another.

Q2. What is the difference between IPv4 and IPv6?

Answer in 2 lines.

Ans: 1. **IPv4** uses a **32-bit address** and supports about **4.3 billion unique IP addresses**.

2. IPv6 uses a **128-bit address**, supports a much larger number of IP addresses, and provides better security and efficiency.

Q3. Run nslookup google.com on your system and write down the IP address you get.

Ans: Addresses: 2404:6800:4000:100e::71

2404:6800:4000:100e::66

2404:6800:4000:100e::64

2404:6800:4000:100e::8a

142.251.106.113

142.251.106.102

142.251.106.139

142.251.106.138

142.251.106.101

142.251.106.100

Q4. What do these HTTP status codes mean: 200, 301, 404, 500, 403?

- Ans: 1. **200 OK** – The request was successful, and the server returned the requested data.
2. **301 Moved Permanently** – The requested page has been permanently moved to a new URL.
3. **404 Not Found** – The requested page or resource could not be found on the server.
4. **500 Internal Server Error** – The server encountered an unexpected error while processing the request.
5. **403 Forbidden** – The server understands the request but refuses to give access to the resource.

Q5. Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

Ans: The DNS (Domain Name System) hierarchy is organized into different levels to help locate websites on the internet.

1. **Root (.)** – The highest level in the DNS hierarchy. It is represented by a dot (.) but is usually hidden.
2. **TLD (Top-Level Domain)** – The last part of the domain name. In www.google.com, **.com** is the TLD.
3. **Domain** – The main name of the website. In www.google.com, **google** is the domain.
4. **Subdomain** – A part added before the domain name to organize or access specific services. In www.google.com, **www** is the subdomain.

Example:

- **Root:** . (hidden)

- **TLD:** com
- **Domain:** google
- **Subdomain:** www

Session 4

Q1. List all 6 phases of SDLC in order with one line about each.

Ans: 1. **Requirement Analysis** – Gather and analyze the customer's needs and project requirements.

2. **System Design** – Create the system architecture, database, and overall design of the software.

3. **Development (Coding)** – Write the program code based on the approved design.

4. **Testing** – Test the software to find and fix bugs and ensure it works correctly.

5. **Deployment** – Release the software so that users can start using it.

6. **Maintenance** – Update, improve, and fix issues in the software after deployment.

Q2. Write 3 differences between Agile and Waterfall models.

Ans: 1. **Agile** is flexible and allows changes during development, whereas **Waterfall** follows a fixed sequence and changes are difficult.

2. **Agile** delivers the software in small, continuous updates, whereas **Waterfall** delivers the complete software at the end of the project.

3. **Agile** involves regular customer feedback throughout the project, whereas **Waterfall** usually involves customer feedback only after development is completed.

Q3. Name the 3 main Scrum roles and write one line about the responsibility of each.

Ans: 1. **Product Owner** – Defines the product requirements and prioritizes the work to deliver maximum value.

2. **Scrum Master** – Guides the Scrum team, removes obstacles, and ensures Scrum practices are followed.

3. **Development Team** – Designs, develops, tests, and delivers the working software during each sprint.

Q4. What is a Sprint in Scrum? Mention its typical duration.

Ans: A **Sprint** is a fixed period during which the Scrum team develops and delivers a working part of the software.

Typical Duration: A Sprint usually lasts **1 to 4 weeks**, with **2 weeks** being the most common duration.

Q5. What do these Jira board columns mean: To Do, In Progress, Done, Backlog?

- Ans: 1. **To Do** – Tasks that are planned and ready to be started.
2. **In Progress** – Tasks that are currently being worked on.
3. **Done** – Tasks that have been completed successfully.
4. **Backlog** – A list of all pending tasks, features, or issues that are planned for future work.

Session 5

Q1. Write the Git command for each: initialize a repo, stage all files, commit with a message, push to remote, pull latest changes.

- Ans: 1. **Initialize a repository** - `git init`
2. **Stage all files** - `git add`
3. **Commit with a message** - `git commit -m "Your commit message"`
4. **Push to the remote repository** - `git push origin main`
5. **Pull the latest changes** - `git pull origin main`

**Q2. What is the difference between git fetch and git pull?
Answer in 2 lines.**

- Ans: 1. **git fetch** downloads the latest changes from the remote repository but does not merge them into your local branch.
2. **git pull** downloads the latest changes and automatically merges them into your current local branch.

Q3. Create a new GitHub repository, push any small text file from your local system, and share the repo link.

Ans:

https://github.com/Purvang07/04July_Purvang_SE/blob/main/hello.txt

Q4. What is the difference between a Fork and a Pull Request? Answer in 2 lines.

Ans: 1. A **Fork** is a copy of someone else's GitHub repository in your own GitHub account.

2. A **Pull Request** is a request to merge your changes from a fork or branch into the original repository.

Q5. Create a new branch named feature-test, make one commit on it, and merge it back to the main branch. Write the commands you used.

Ans: git checkout -b feature-test

echo "This is a test file." > test.txt

git add .

git commit -m "Added test file"

git checkout main

git merge feature-test

These are the commands used to create the feature-test branch, make a commit, switch back to the main branch, and merge the changes.

